ModelIt!

Teacher's Guide

Why Do Invasive Species Win?

Modeling Ecological Competition, Niche Displacement, and Biodiversity Collapse

Quick Reference

Grade Level	Grade 11L2
NGSS Standard	HS-LS2-2, HS-LS2-6, HS-LS4-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-2, HS-LS2-6, HS-LS4-5

Use mathematical representations to support and revise explanations based on evidence about factors affecting biodiversity and populations in ecosystems of different scales. Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem. Evaluate the evidence supporting claims that changes in environmental conditions may result in increases in the number of individuals of some species, the emergence of new species over time, and the extinction of other species.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use mathematical representations of population dynamics, competitive exclusion, and resource partitioning to model how invasive species displace native communities and to predict thresholds at which management interventions succeed or fail.
Disciplinary Core Idea	**LS2.A Interdependent Relationships in Ecosystems / LS2.C Ecosystem Dynamics / LS4.D Biodiversity and Humans** — Students model how species interactions maintain ecosystem stability, how the introduction of invasive competitors disrupts these relationships, and how biodiversity loss from invasion degrades ecosystem services that humans depend on.
Crosscutting Concept	**Cause and Effect / Stability and Change** — Students analyze the cause-and-effect chains through which invasive species destabilize ecosystems, identifying the feedback loops and tipping points that determine whether ecosystems can absorb the disturbance or collapse to a degraded state.

CAST Testing Alignment

MULTIPLE CHOICE:

Invasive species have contributed to 60% of all documented extinctions and cost $423 billion annually worldwide. The invasion curve shows that management costs increase by 10-100x for each stage of invasion (prevention → early detection → containment → long-term management). Based on your model, which strategy provides the GREATEST return on investment for protecting biodiversity?

CONSTRUCTED RESPONSE:

Using your model, explain why a healthy, diverse ecosystem with dozens of native species can collapse within a decade when a single invasive species is introduced. Reference competitive exclusion, enemy release, positive feedback loops, and the concept of cascading extinction in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Invasive species represent one of the most powerful and underappreciated drivers of global biodiversity loss. Understanding biological invasions requires modeling the intersection of population ecology, competition theory, evolutionary biology, and human-mediated transport to explain why organisms that evolved on one continent can devastate ecosystems on another — and why the scale of the problem is accelerating with globalization.

The Scale of the Crisis

Invasive species are the second leading cause of species extinction worldwide, surpassed only by habitat destruction. They have contributed to 60% of all documented plant and animal extinctions since 1500. The rate of new introductions has increased by 40% since 1970, driven by expanding international trade, travel, and shipping. The global economic cost of invasive species is estimated at $423 billion annually — including agricultural losses, infrastructure damage, control efforts, and ecosystem service degradation. In the United States alone, invasive species cause an estimated $120 billion in annual damages. The problem is accelerating: climate change is expanding the geographic range in which warm-adapted invasive species can establish, while simultaneously stressing native species and reducing their competitive ability.

Why Invaders Win: The Competitive Asymmetry

Invasive species succeed through a combination of traits and circumstances that create a fundamental competitive asymmetry. The enemy release hypothesis explains the most important factor: in their native range, species are controlled by co-evolved predators, parasites, herbivores, and diseases that limit population growth. When transported to a new environment, invaders leave these natural enemies behind while native species have not evolved defenses against the newcomer. This asymmetry is compounded by the traits common to successful invaders: rapid reproduction, short generation times, broad environmental tolerances, generalist diets, high dispersal ability, and phenotypic plasticity. Burmese pythons in the Everglades, for example, face no predators, produce 50-100 eggs per clutch, tolerate a wide range of habitats, and eat virtually anything — they have reduced mammal populations by 90-99% in some areas.

Invasion Dynamics and Tipping Points

Biological invasions follow a characteristic pattern with critical thresholds. The introduction phase is the initial arrival of a small founding population. Most introductions fail — perhaps 10% of introduced species establish self-sustaining populations (the 'tens rule'). But those that survive the establishment phase enter a lag phase where the population grows slowly and remains localized, often for years to decades. If conditions allow, the population eventually crosses the establishment threshold — a critical density above which growth becomes exponential and self-sustaining. The subsequent spread phase is characterized by rapid range expansion, resource monopolization, and native species displacement. Critically, many invasive species create positive feedback loops that accelerate their own spread: invasive grasses increase fire frequency (which kills native trees and creates more grassland), invasive mussels filter-feed plankton (reducing food for native species while increasing water clarity that favors invasive aquatic plants), and invasive predators reduce native prey (releasing invasive plants from herbivory).

Management Strategies and the Invasion Curve

The invasion curve illustrates that management cost-effectiveness decreases exponentially with invasion stage. Prevention — biosecurity screening at borders, trade regulation, pathway risk analysis — costs pennies per species prevented and is 100-1,000 times more cost-effective than any later intervention. Early detection and rapid response — systematic monitoring, eDNA sampling, citizen science networks — can eradicate incipient invasions when populations are small and localized. Once invasion passes the establishment threshold, eradication becomes progressively more difficult and eventually impossible for widespread species. Long-term management — biocontrol, mechanical removal, habitat restoration — can reduce invasive species impacts but rarely eliminates them. Biocontrol introduces a natural enemy from the invader's native range; when carefully researched (10-15 years of host-specificity testing), it can provide self-sustaining population control. However, poorly researched

biocontrol has itself created invasive species disasters — the cane toad introduction in Australia being the most infamous example.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify six components of the invasion system: Invasive Species Introduction Rate, Invasive Population Growth Rate, Native Species Competitive Fitness, Resource Availability, Native Biodiversity Index, and Ecosystem Function Integrity — distinguishing the human-mediated introduction from the ecological processes that determine invasion success or failure.

E **Establish Relationships**

Students determine that Invasive Species Introduction Rate initiates the invasion, but Invasive Population Growth Rate (driven by enemy release and reproductive advantage) determines whether the population establishes. As the invasive population grows, it reduces Resource Availability for native species, which declines Native Species Competitive Fitness. As natives decline, Native Biodiversity Index drops, degrading Ecosystem Function Integrity.

V **Visualize & Model**

Students build a computational model showing the competitive dynamics between invasive and native populations — visualizing the establishment threshold, the exponential growth phase, and the cascading collapse of native biodiversity.

E **Evaluate Outcomes**

Students run early detection, established invasion, and managed ecosystem scenarios to identify the critical population threshold that determines whether intervention succeeds and quantify the cascade of biodiversity loss at each invasion stage.

R **Revise & Extend**

Students add Disease and Parasite Introduction, Habitat Modification Feedback, or Genetic Hybridization to explore how invasive pathogens, ecosystem engineering by invaders, and genetic swamping create additional and often more devastating invasion pathways.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.

At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"This forest took centuries to grow. This vine took a decade to smother it. Invasive species are winning a war that most people don't even know is happening."*

Watch For

Show before-and-after photos of the same landscape — one with diverse native habitat, one overtaken by a monoculture of invasive species (kudzu, water hyacinth, or lionfish). Ask: What happened to everything that used to live here?

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling why evolution's finest products — species refined over millions of years — lose to newcomers that arrived yesterday. The answer will change how you think about ecosystems."*

Watch For

Have students read objectives. Pre-teach the enemy release hypothesis with an analogy: imagine a champion boxer who fights every week vs. a boxer who hasn't fought in years. Now the champion arrives in a ring where nobody has ever seen boxing. That's the invasive species advantage. Quick-write: What makes a species invasive versus just non-native?

Slide 3: Big Question

3 min

Say: *"Native species are perfectly adapted to their environment — the result of millions of years of natural selection. So why does a snake from Burma eliminate 99% of mammals in the Everglades in just 20 years?"*

Watch For

Present the Burmese python case study in 60 seconds: introduced through pet trade, no predators, 50-100 eggs per clutch, eats everything. Mammal populations declined 90-99%. Ask: What would it take to remove them? Is it even possible?

Slide 4: LEVER Framework

2 min

Say: *"We're modeling a competition where one side has been fighting this battle for millions of years and the other side just got all the cheat codes unlocked. The invader is in the game — without the rules."*

Watch For

Preview each LEVER step. Draw the invasion curve on the board: flat (introduction), slight rise (establishment), near-vertical (spread), plateau (dominance). Mark the intervention window: 'This narrow zone is where we can win. After this line, we're managing, not solving.' Let that land.

Slide 5: Activity 1: Components

8 min

Say: *"One human action — moving species where they don't belong. Five ecological responses. But every response tips the playing field further toward the invader."*

Watch For

Guide sorting of external vs. internal components. Only Invasive Species Introduction Rate is truly external — it depends on human trade, travel, and transport decisions. Discuss why Invasive Population Growth Rate is internal: once introduced, the ecological dynamics (enemy release, resource abundance) drive growth without further human input.

Slide 6: Activity 2: Connections

8 min

Say: *"The invader grows. Resources shift. Natives decline. The ecosystem changes in ways that help the invader more. It's a positive feedback spiral — and there's a point of no return."*

Watch For

Help students map the competitive dynamics. Identify the critical positive feedback loop: invader population grows → modifies habitat → environment becomes more favorable for invader → population grows faster. Find the tipping point where native decline becomes self-reinforcing (loss of pollinators reduces native plant reproduction, which reduces native herbivore food, which reduces native predator prey).

Slide 7: Activity 3: Simulation

10 min

Say: *"Three windows into the invasion: catch it early, catch it late, or try to manage what you missed. The difference in outcomes will stun you."*

Watch For

Students predict before running simulations. Key focus: the dramatic difference in outcomes between early detection (population below threshold) and established invasion (above threshold). In the early scenario, a small effort succeeds. In the established scenario, a massive effort barely contains. Compare the cost-effectiveness ratio across scenarios.

Slide 8: Discoveries

5 min

Say: *"Your model just revealed the invasion paradox: we spend billions managing invasions we could have prevented for millions. The most powerful tool in conservation isn't a trap or a poison — it's a border inspection."*

Watch For

Lead discussion on the invasion curve economics: $1 in prevention = $100-$1,000 in management. Connect to real-world examples: Asian carp approaching the Great Lakes, brown tree snakes in Guam (eliminated 10 of 12 native bird species), zebra mussels in the Great Lakes ($1 billion/year in damages). Discuss: Should we spend more on preventing introductions or managing existing invasions?

Slide 9: STEM Challenge

12 min

Say: *"Fifteen endangered species depend on this preserve. Three invasions slipped through in the last decade. Design the system that catches the next one before it's too late."*

Watch For

Groups design comprehensive monitoring and rapid response systems including technology selection (eDNA, remote sensing, camera traps), monitoring network design, population threshold triggers, response protocols, and partnership frameworks. Present as proposals to a conservation board protecting endangered species.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Invasive Species Introduction Rate
- **INTERNAL:** Invasive Population Growth Rate, Native Species Competitive Fitness, Resource Availability, Native Biodiversity Index, Ecosystem Function Integrity

Relationships

- **Invasive Species Introduction Rate to Invasive Population Growth Rate:** POSITIVE (+) — More frequent introductions increase the probability that at least one founding population will survive initial environmental filtering and reach the establishment threshold. Multiple introduction events also increase genetic diversity in the invasive population, improving its adaptive potential.
- **Invasive Population Growth Rate to Resource Availability:** NEGATIVE (-) — As the invasive population grows, it consumes, degrades, or monopolizes resources that native species also depend on. Invasive plants shade out native seedlings, invasive predators deplete native prey, and invasive filter-feeders strip nutrients from the water column.
- **Resource Availability to Native Species Competitive Fitness:** POSITIVE (+) — When resources are abundant, native species can maintain healthy population sizes and reproductive success even in the presence of competitors. As resources decline due to invasive consumption, native species face food scarcity, habitat loss, and increased competition stress.
- **Native Species Competitive Fitness to Native Biodiversity Index:** POSITIVE (+) — When native species maintain competitive fitness, populations remain viable and species richness is preserved. As fitness declines, populations drop below minimum viable sizes, local extinctions occur, and biodiversity decreases — with specialist species disappearing first and generalists persisting longest.
- **Native Biodiversity Index to Ecosystem Function Integrity:** POSITIVE (+) — Higher biodiversity supports more complete ecosystem function — pollination networks, decomposition cycles, nutrient transport, predator-prey balance, and disease regulation all depend on having sufficient species to fill functional roles. As biodiversity declines, ecosystem services degrade in a nonlinear pattern where the loss of keystone species triggers disproportionate functional collapse.

Key Discoveries

- Invasive species are the second leading cause of species extinction globally (after habitat destruction) and have contributed to 60% of all documented plant and animal extinctions — costing the global economy an estimated $423 billion annually in damages and control efforts
- The 'invasion curve' demonstrates that management cost-effectiveness declines exponentially with invasion stage — spending $1 on prevention or early detection avoids $100-1,000 in later eradication or damage control costs
- Invasive species create positive feedback loops: as they modify habitat (e.g., invasive grasses increasing fire frequency), they make the environment more favorable for themselves and less favorable for natives, accelerating displacement in a self-reinforcing cycle
- Island ecosystems and isolated habitats (lakes, mountaintops) are disproportionately vulnerable because their native species evolved without exposure to aggressive competitors or predators — 75% of all documented extinctions have occurred on islands

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"All non-native species are invasive"	Only a small fraction of introduced non-native species become invasive — the 'tens rule' estimates that roughly 10% of introduced species establish self-sustaining populations, and roughly 10% of those become invasive (causing ecological or economic harm). Many non-native species coexist with native communities without significant impact. The distinction between non-native (present but benign) and invasive (present and harmful) is critical for directing limited management resources toward genuine threats.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Nature will eventually reach a new balance with invasive species"	While ecosystems can eventually incorporate non-native species into new community structures, this 'new balance' typically involves dramatic biodiversity loss, reduced ecosystem function, and the permanent extinction of species that cannot adapt or relocate. The new equilibrium may be stable but drastically impoverished — like the Guam forest that functions without birds or the Great Lakes fishery restructured around invasive species. Evolution can produce co-adaptation, but it operates on timescales of thousands to millions of years — far longer than the decades over which invasions cause extinction.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We can always eradicate invasive species if we try hard enough"	Eradication is only feasible for small, localized populations detected early — typically below the establishment threshold. Once an invasive species is widespread, eradication is virtually impossible for all but the smallest and most isolated ecosystems (small islands). The history of invasive species management is littered with failed eradication attempts that consumed billions of dollars. This is not a failure of effort — it is a mathematical reality: organisms that reproduce exponentially, disperse widely, and have no natural enemies cannot be removed faster than they reproduce once populations are large. The model makes this clear: prevention and early detection are not 'better' strategies — they are the only strategies that work.